



Respiration — Yeast Breathes Too!



The Fluffy Naan Mystery

Kabir and his mother visit a dhaba and wonder about the tiny pockets in Amritsari Kulchas.

Mother's secret: "A million tiny invisible bakers blew little air balloons inside the dough!"

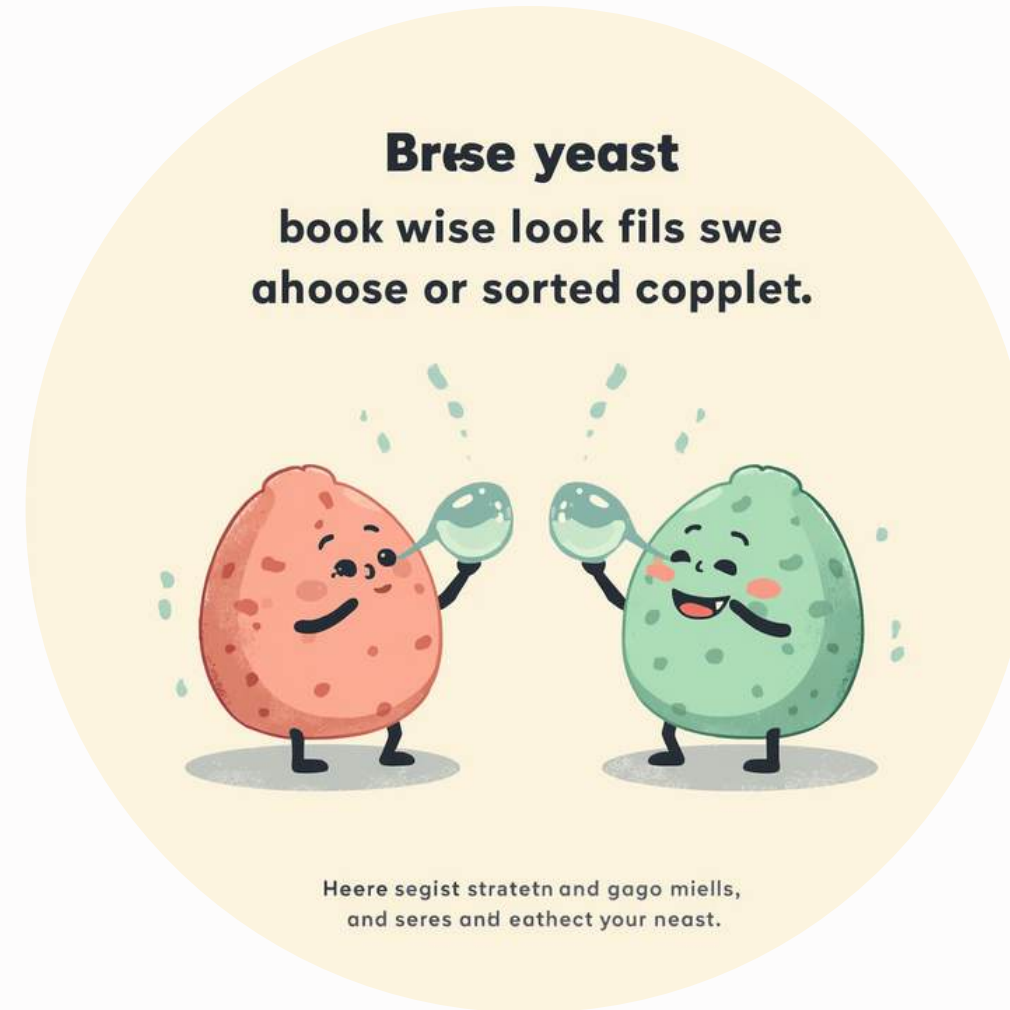


Meet Yeast – Tiny Sleeping Giants

Yeast are tiny, friendly microbes—like sleeping dots that need food and warmth to wake up!

- Yeast are alive but very, very tiny
- They love warm places and sweet treats
- They look like tiny mustard seeds (Rai)!





Yeast's Special Breath

When yeast eats sugar, it wakes up and breathes out invisible gas bubbles. These tiny bubbles help dough rise like little balloons!

The Balloon Bottle Experiment Setup



Warm Water

Not too hot, not too cold — just cozy warm water to wake up the yeast!



Sugar Food

Yeast loves sweet treats! Add a spoonful of sugar for yeast to eat.



Balloon on Top

Stretch a balloon over the bottle to catch the invisible gas yeast breathes out!

Watch what happens when yeast wakes up!

Watch the Balloon Inflate!

Yeast breathes invisible gas that fills
the balloon slowly and steadily.
Watch the magic happen!

The balloon grows bigger over time as
yeast keeps breathing out gas bubbles.
This is how yeast makes dough rise too!



The Yeast Chef Simulation Introduction

Let's be yeast chefs! We will test how yeast reacts to different water and sugar combinations. Get ready with three test bowls labeled Bowl 1, Bowl 2, and Bowl 3 for our experiment!



Three Test Bowls

Results Comparison



Bowl 1: Cold Water + Sugar

No movement at all! The yeast stays asleep in cold water. Too chilly to wake up!



Bowl 2: Warm Water Only

Weak tiny bubbles appear. Yeast wakes up but is hungry without sugar to eat!



Bowl 3: Warm Water + Sugar

Big bubbles everywhere! Happy yeast eats sugar and breathes lots of invisible gas!



Finding the Happy Warm Zone

Yeast needs just the right warm zone to wake up and breathe! Not too cold, not too hot — drag the slider to find the best temperature for yeast.

Quick Check!

What Wakes Up Yeast?



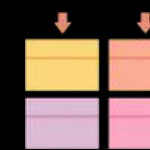
Question 1

What wakes up sleeping yeast? Choose: Warm water, Ice water, or No water?



Question 2

What does yeast breathe out? Choose: Invisible gas, Smoke, or Water?



Question 3

Sort it! Drag sugar and warmth to "Yeast's Food" and invisible gas to "Yeast's Breath."

Why Yeast is Our Kitchen Helper



Tiny Friendly Microbe

Yeast is a tiny living helper we cannot see with our eyes, but it does big work!



Wakes Up Happy

Yeast loves warm water and sweet sugar to wake up and start working.



Makes Dough Fluffy

Yeast breathes invisible gas that fills dough with soft, fluffy air pockets!



Bakery Detective

Your Home Quest

Find 5 tiny bubble windows in bread, idli, or naan at home. These air pockets are the secret breath of yeast!

Look closely at your rotis, parathas, and pav. Can you spot where the tiny yeast helpers blew their invisible bubbles?



Thank You!

See You Next Time!

Thank you for discovering yeast's secret breath with us!
Keep exploring the tiny helpers in your kitchen.



PLANRS Academy



Yeast Dance Party!



Heat yeast dance party.



Yeast loves to dance when warm sugar is around!